

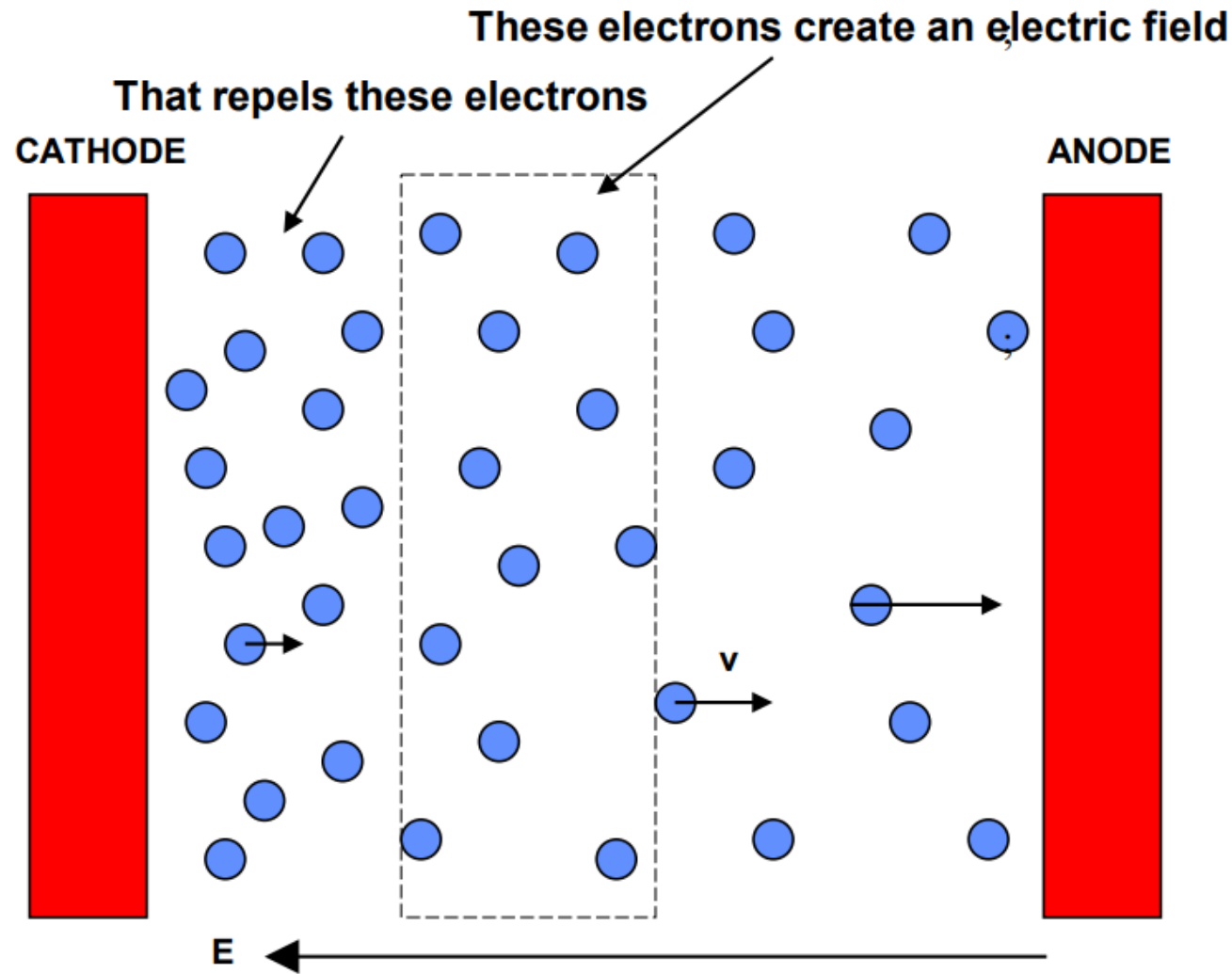
Lecture 13

Tuesday 7/21/26

Sami Lewis

US Particle Accelerator School
Accelerator Physics – Summer 2026

Diode with space charge



Child-Langmuir Law (non-relativistic)

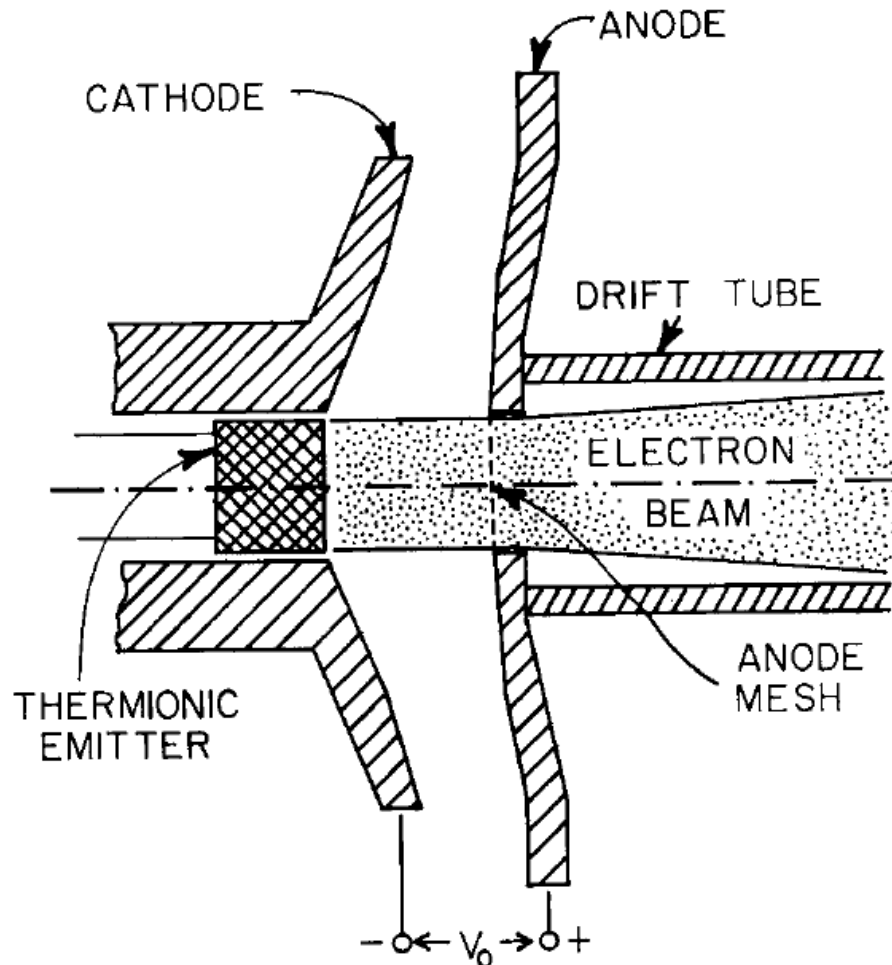
$$J = \frac{4}{9} \epsilon \left(\frac{2q}{m} \right)^{1/2} \frac{V_0^{3/2}}{d^2}$$

<https://indico.cern.ch/event/73157/contributions/2076035/attachments/1039879/1481922/Scrivens-web.pdf>

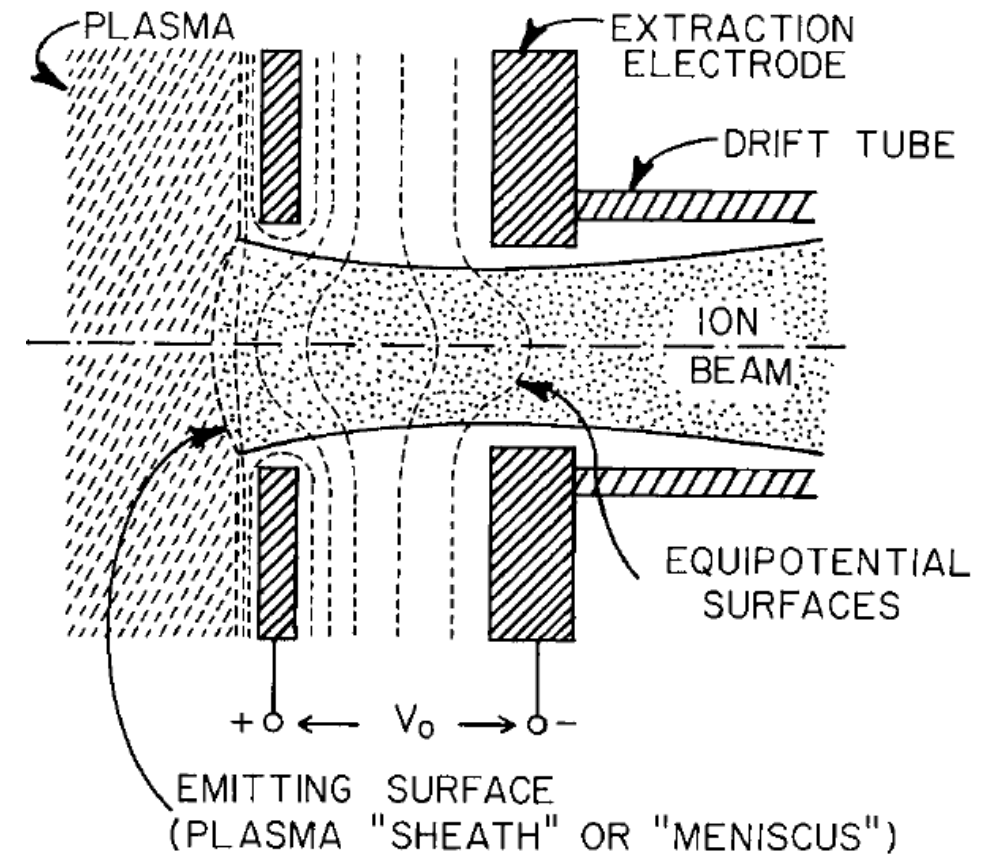
Adding complexity to diode picture

M. Reiser "Theory and Design of Charged Particle Beams"

Electron "diode"

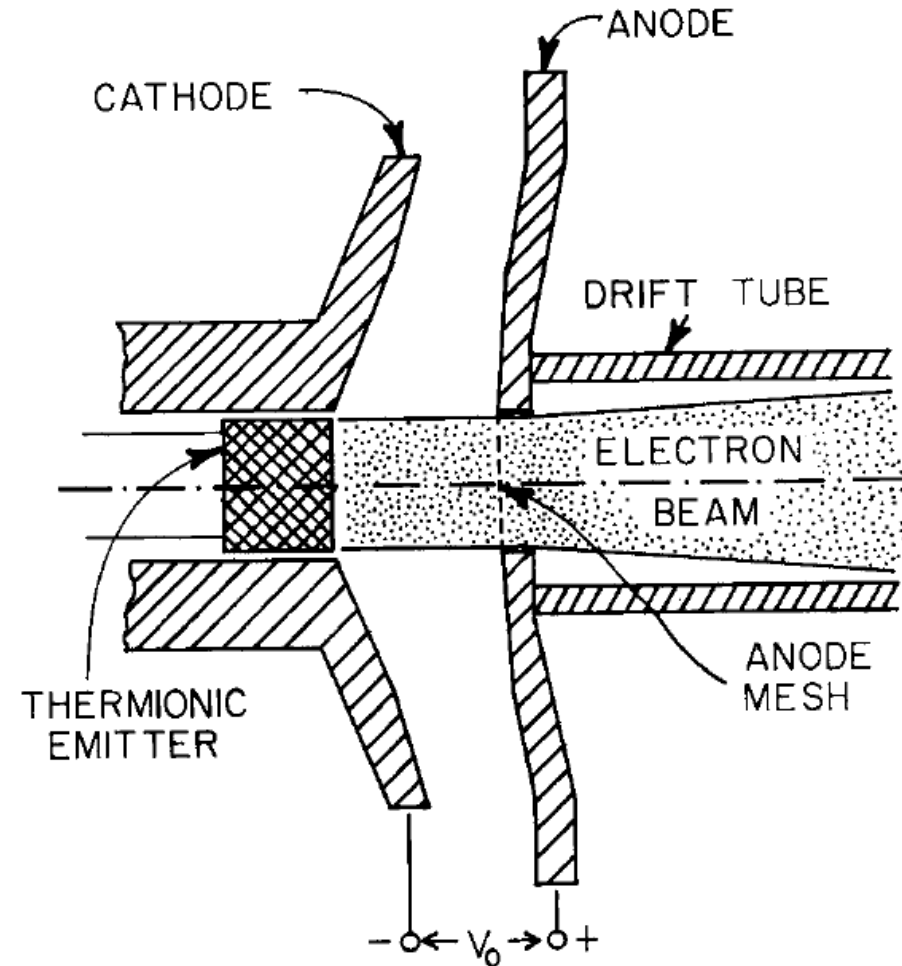


Ion "diode"



Emittance/beam size

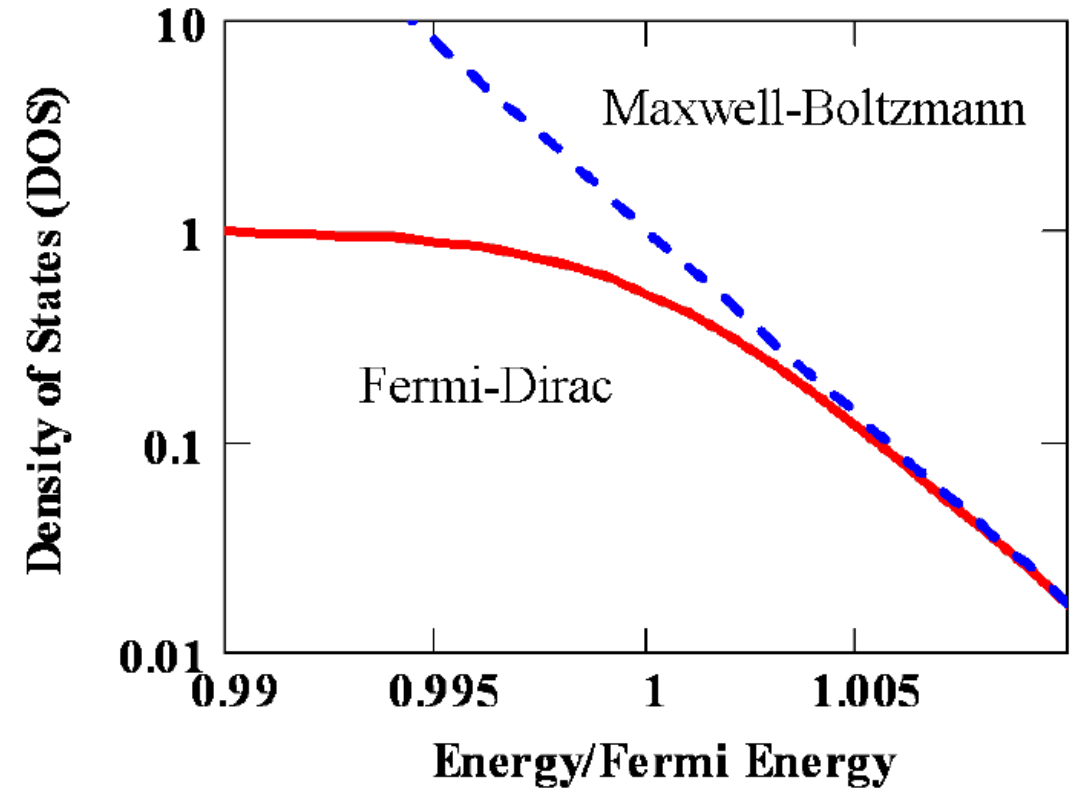
- Even in simple example, can see features related to beam emittance and size
- E.g. anode mesh: reduces transverse fields at the aperture that would cause defocusing
- Also influenced by emitter geometry, more on this later



Electron sources

Electron emission overview

- Electrons can be removed from some surface
- Three mechanisms: thermionic emission, field emission, and photo emission
- Lots of electrons inside a material, many more than emitted
- Electrons are fermions, obey Fermi-Dirac statistics
- Statistical distribution of particles in energy phase space (N particles, k energy levels)



https://uspas.fnal.gov/materials/08UCSC/Lecture2_EmissionStatisticsCathodeEmittance.pdf

Dispersion vector form

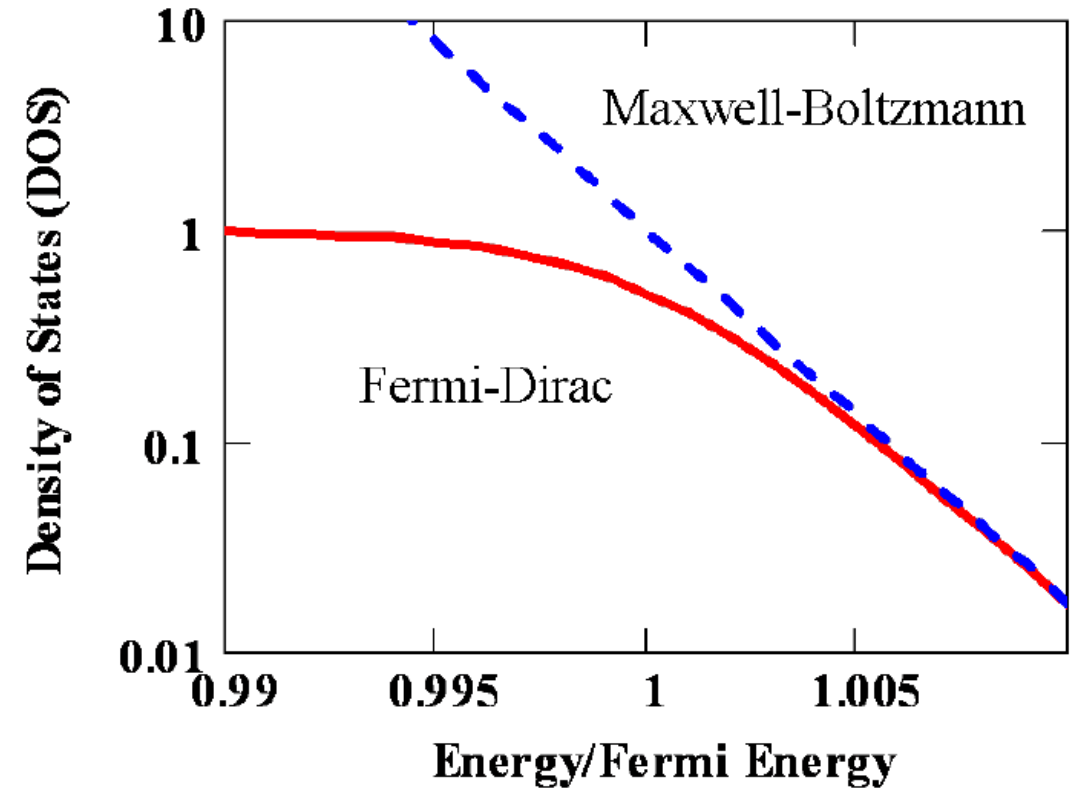
Fermi-Dirac: $f_{FD} = \frac{1}{1 + e^{(E-E_F)/k_B T}}$

Maxwell-Boltzmann: $f_{MB} = \frac{1}{e^{E/k_B T}}$

Fermi level

Boltzmann's constant

Temperature



At high energies tails, both look similar

https://uspas.fnal.gov/materials/08UCSC/Lecture2_EmissionStatisticsCathodeEmittance.pdf

Electron emission overview

- Thermionic: heat materials, electrons gain enough energy to overcome barrier
- Photoemission: excite electron using input photon, enough energy to overcome barrier
- Field emission: lower potential barrier, electrons tunnel
- Can have combinations

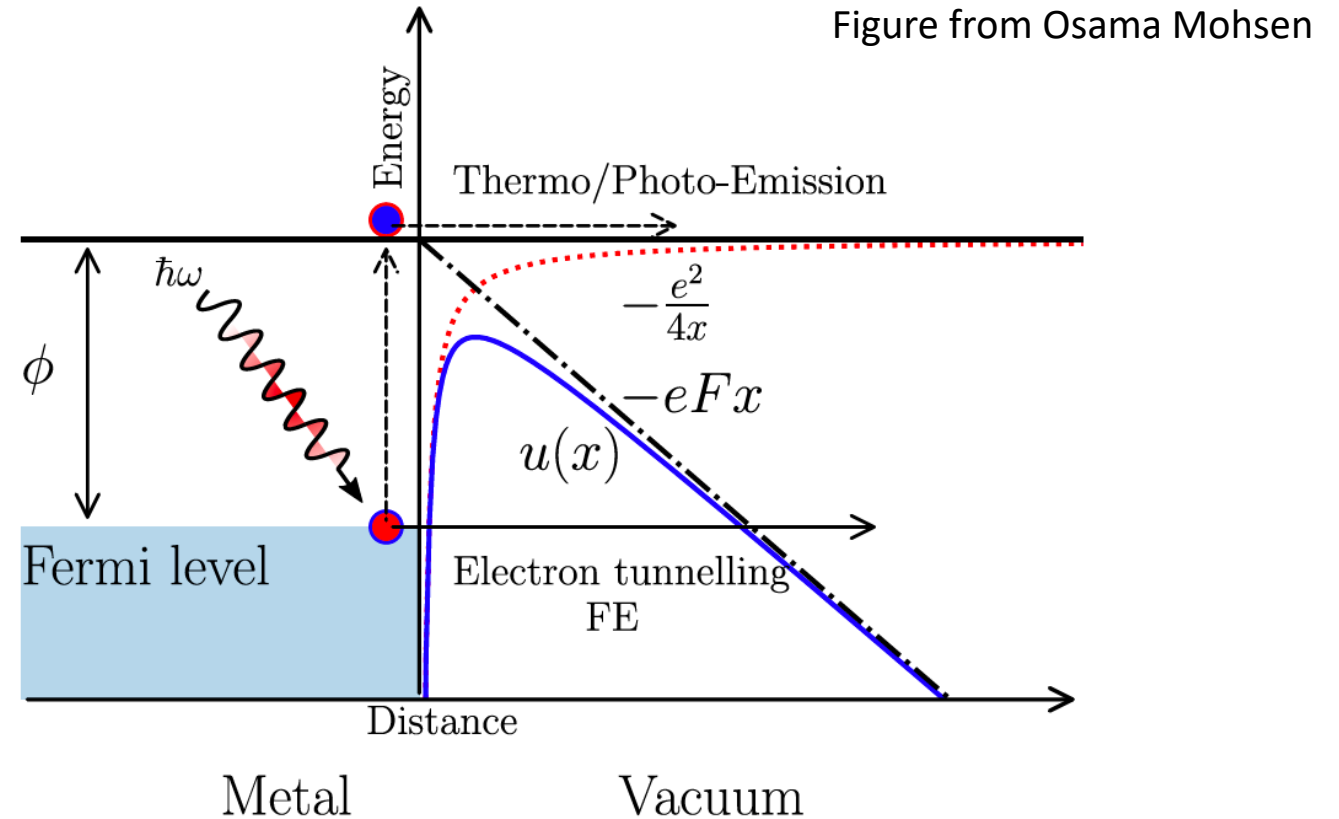


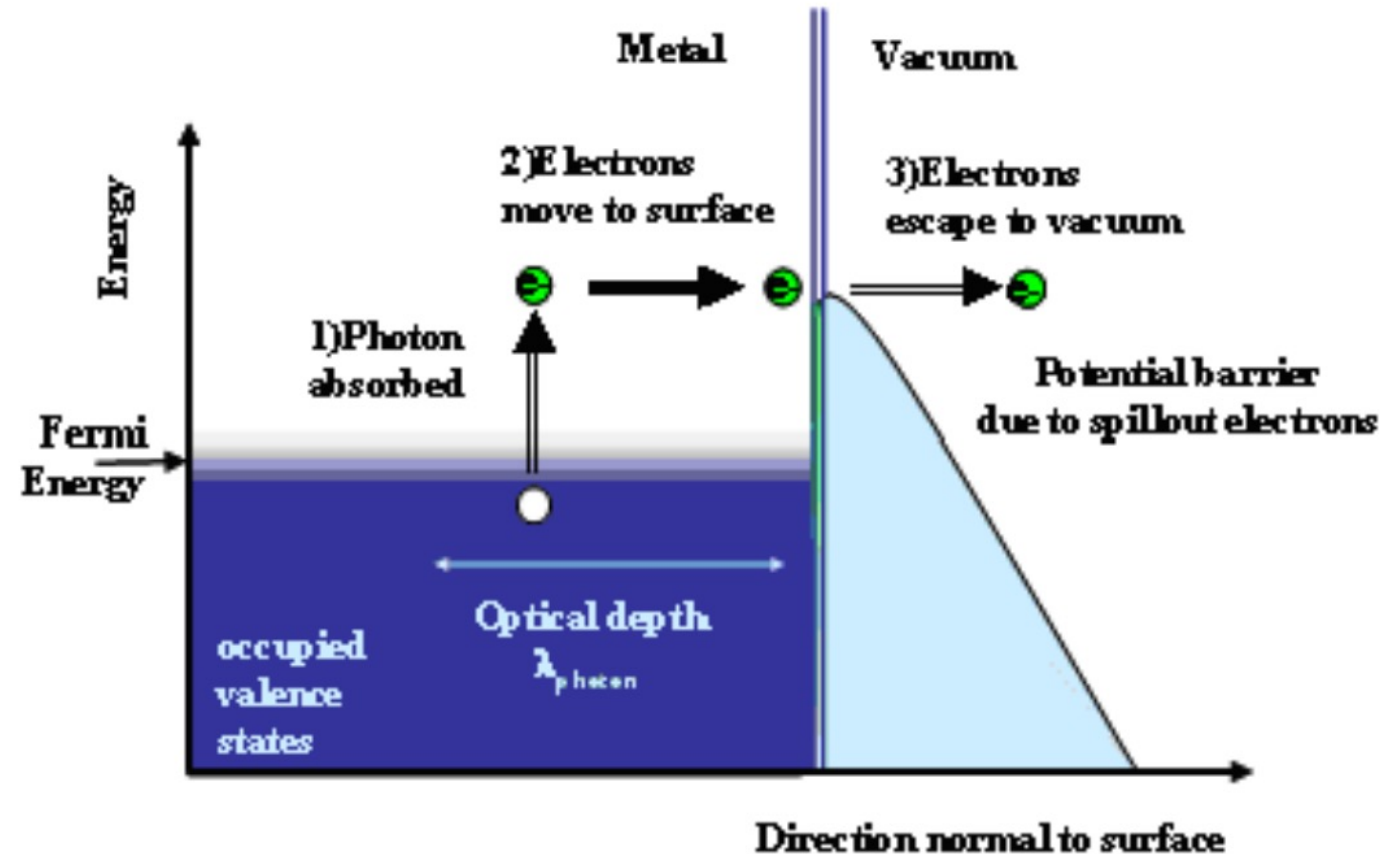
Figure 2.2: Schematic of the emission in case of the thermionic, photo, and field emissions. The strong electric field of strength F lowers the effective vacuum potential $u(x)$ (blue trace) by an amount $-eFx$ (black dashed trace) allowing electrons to tunnel through the potential barrier in case of the Field Emission. In the other cases, electrons need to gain the energy difference to overcome the barrier (dotted red trace) which is provided by electromagnetic radiation or heating for photo and thermionic emission, respectively.

Thermionic emission

- See written class notes

Photoemission

- Spicer 3 step model of photoemission:
 1. Excitation from bound state
 2. Transport to surface
 3. Escape to vacuum
- Some penetration depth of incident radiation into surface
- Can also account for some reduction in barrier from field at surface



Spicer: <https://www.slac.stanford.edu/pubs/slacpubs/6250/slac-pub-6306.pdf>

Photocathodes: <https://cds.cern.ch/record/257133/files/P00020029.pdf>

Photoemission – excitation from bulk

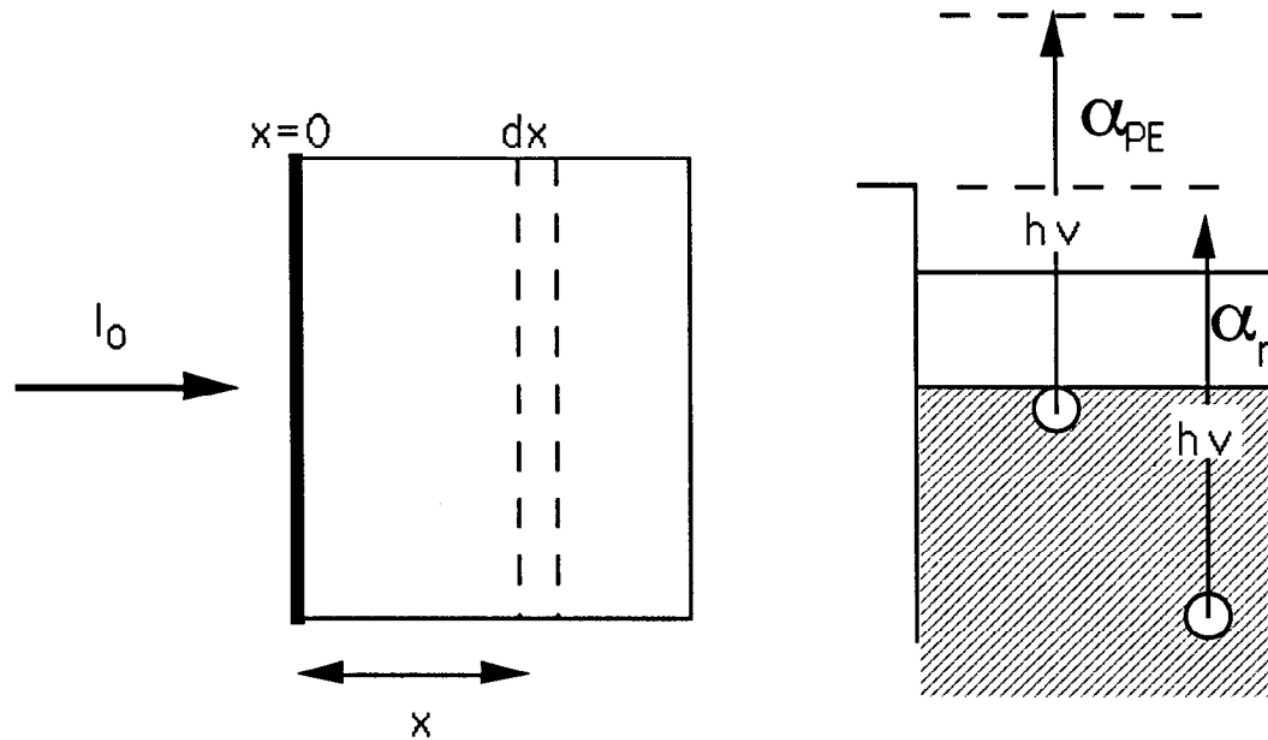


Figure 2. The contribution to the photoemission yield from excitation in the slab of thickness dx depends on the distance x from the surface, where α_{PE} represents the part of the absorption where the electrons are excited above the vacuum level and have a possibility to escape, and α_n is the absorption that does not produce any yield.

Spicer: <https://www.slac.stanford.edu/pubs/slacpubs/6250/slac-pub-6306.pdf>

Photocathodes: <https://cds.cern.ch/record/257133/files/P00020029.pdf>

Photoemission – excitation from bulk

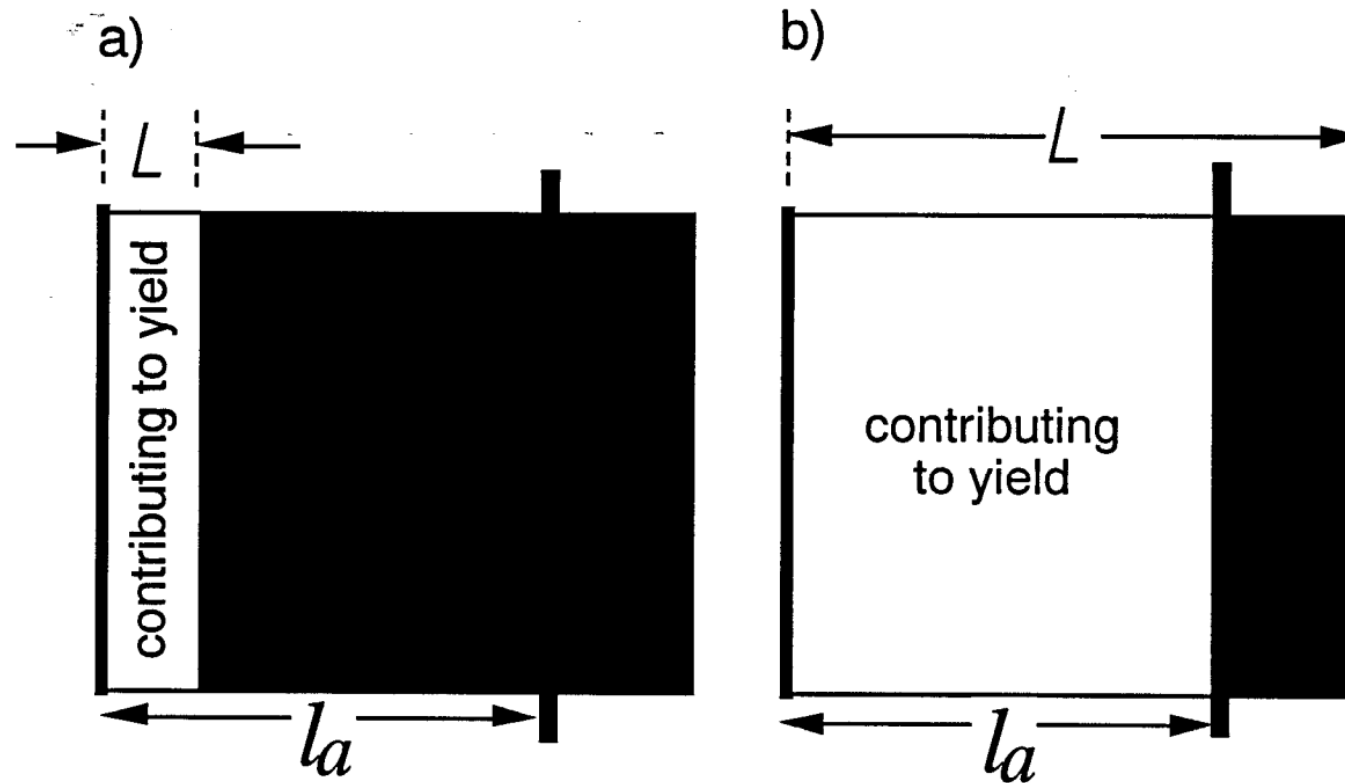


Figure 3. The white volume is the region contributing to the yield. L is the escape length, and l_a is the absorption length. If $L \ll l_a$, then only a small fraction of the photoexcited electron contributes to the yield.

Spicer: <https://www.slac.stanford.edu/pubs/slacpubs/6250/slac-pub-6306.pdf>

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Dowell notes:

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Field emission

- Fowler-Nordheim equation for the field-emitted current, with E in V/m and $\phi = \phi_{work}$ in eV (same as E_0 in written notes, field strength)

$$I_F = \frac{1.54 \times 10^{-6} \times 10^{4.52\phi^{-0.5}} A_e \beta^2 E^2}{\phi} \exp \left(-\frac{6.53 \times 10^9 \phi^{1.5}}{\beta E} \right) \text{ [A]}$$

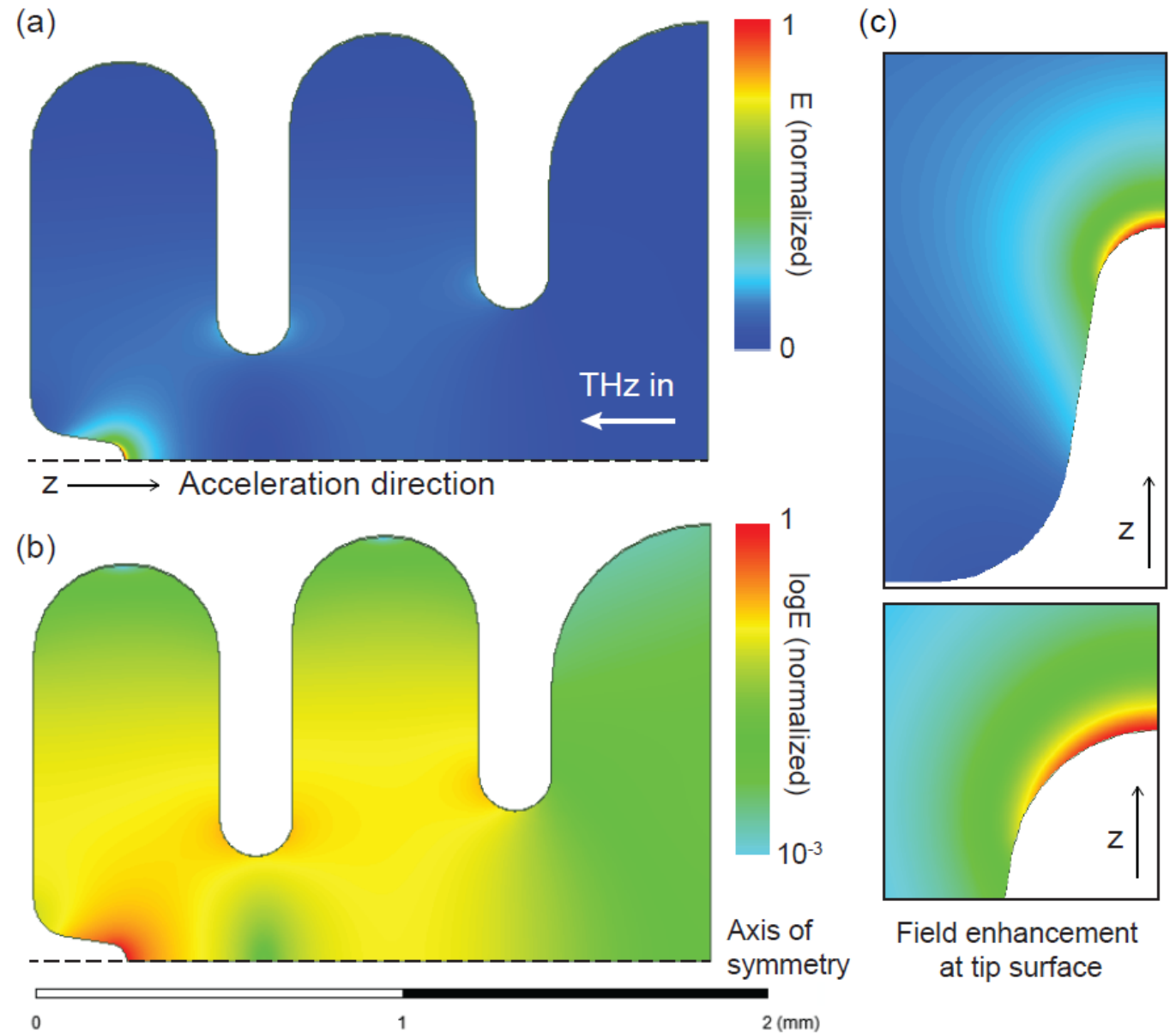
Area of emitter

Field enhancement factor

- Local field enhancement heavily influences emitted current
- Can be intentional: e.g. shaping of a cathode tip
- Can be unintentional: e.g. microscopic surface features, crystal structure

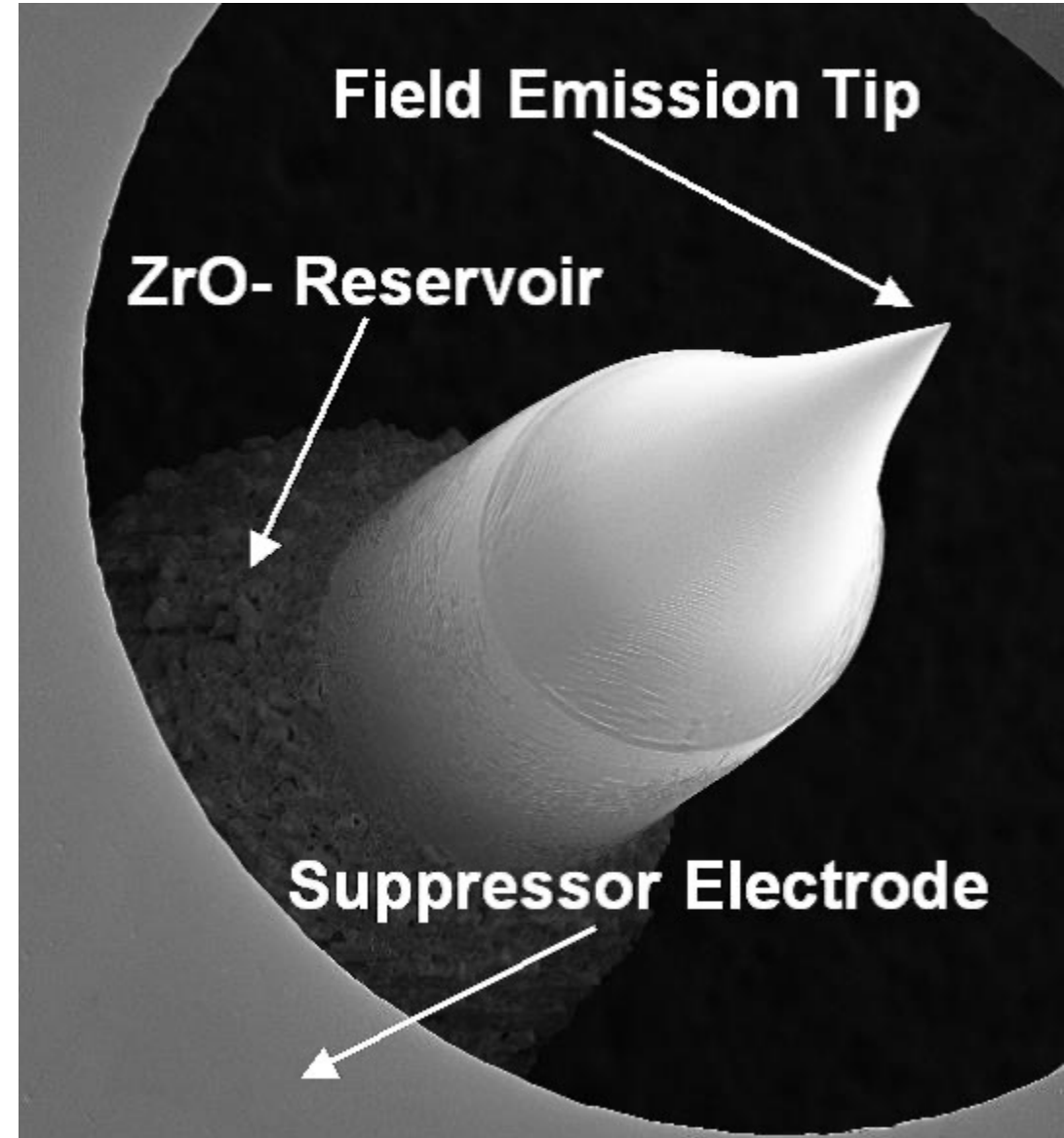
Field emission

- Surface field enhancement: cathode tip shaping
- Size of emitter influences emittance
- Full discussion of analytic expression for emittance: K. L. Jensen *et al.*, J. Appl. Phys. 107, 014903 (2010); <https://doi.org/10.1063/1.3267288>



Field emission

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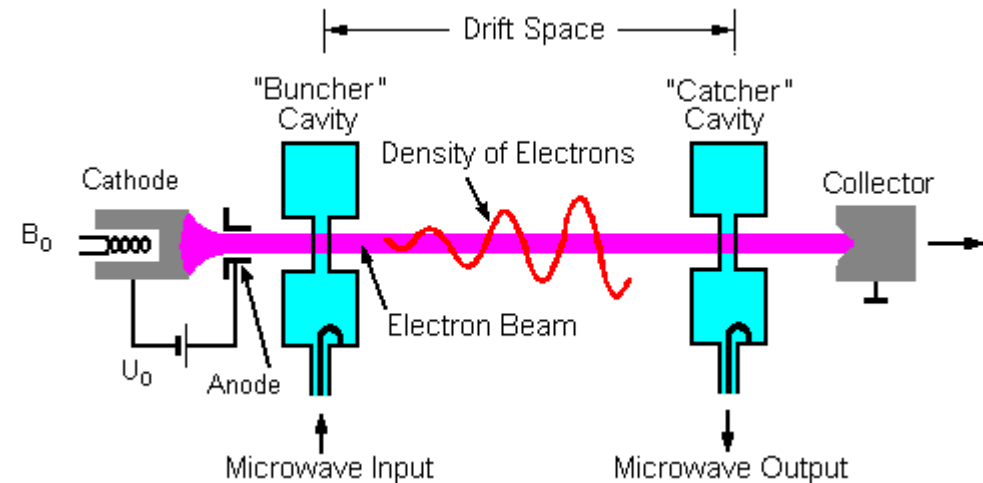
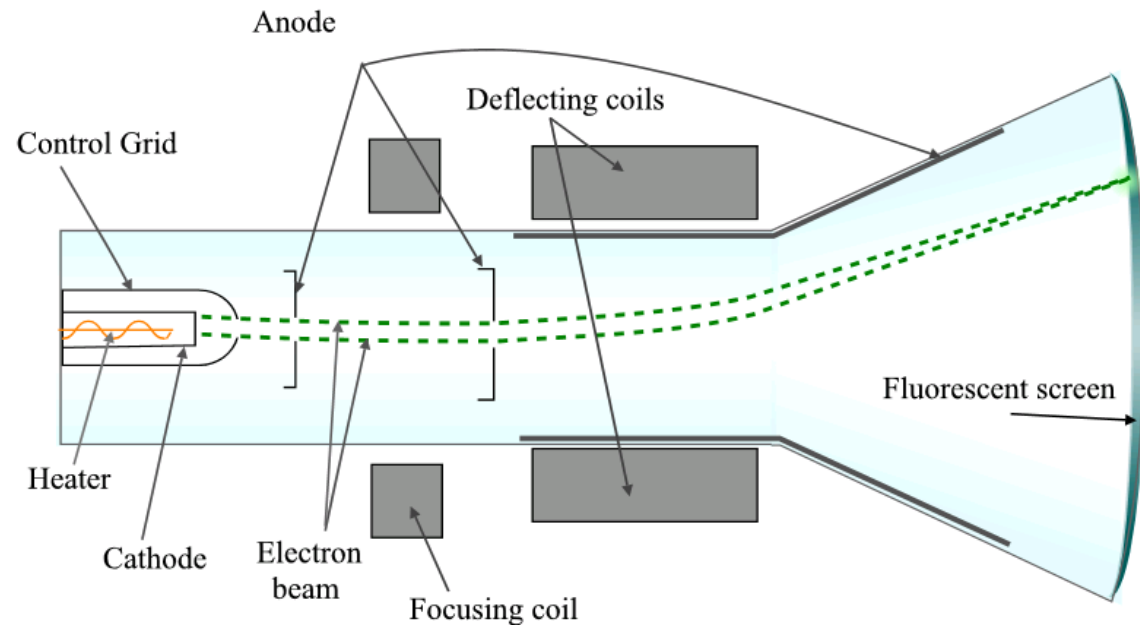


Combining emission types

- Can have combinations of each type
- Example: photo-field emission, where additional high field is applied/tip is shaped to enhance field
- Example: high temperature field emission, requires lower fields since electrons have additional energy from heating (thermionic)

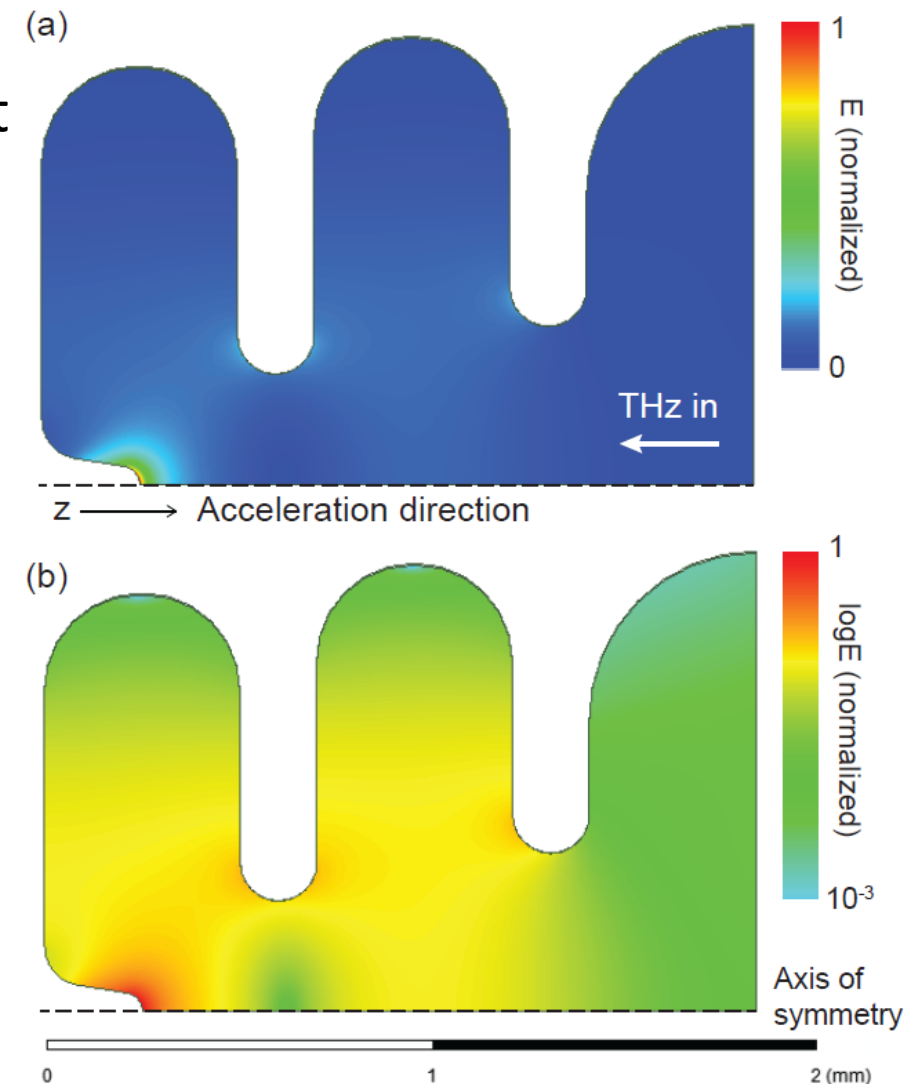
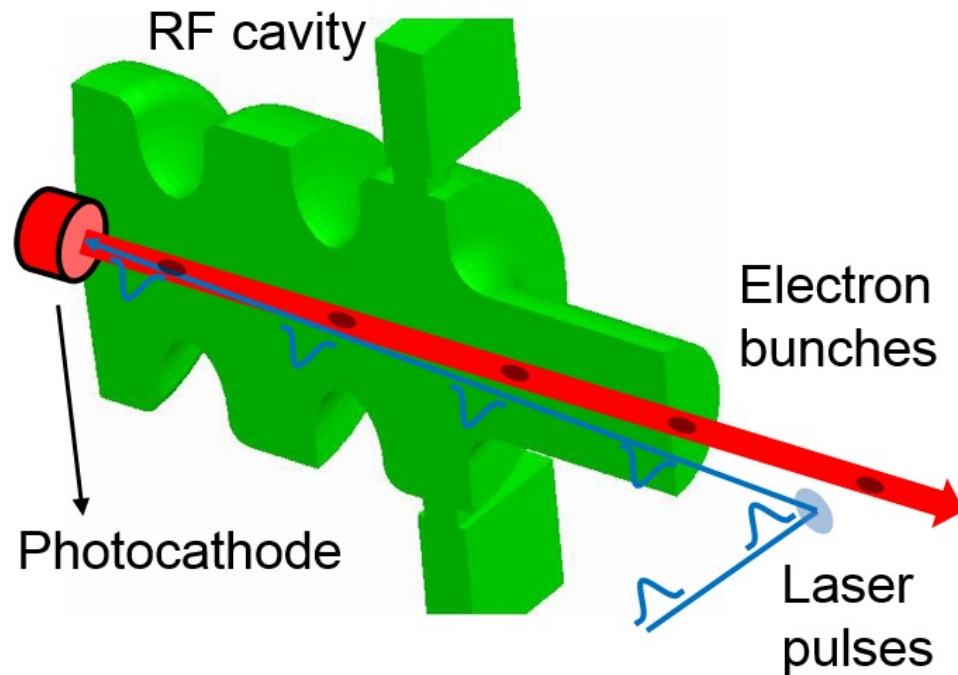
DC electron guns

- Very common, used in a variety of applications, usually with heating
- DC accelerating voltage, like our diode example
- Produces some steady state current of particles
- Example: cathode ray tube
- Common for microwave sources like klystrons



RF electron guns

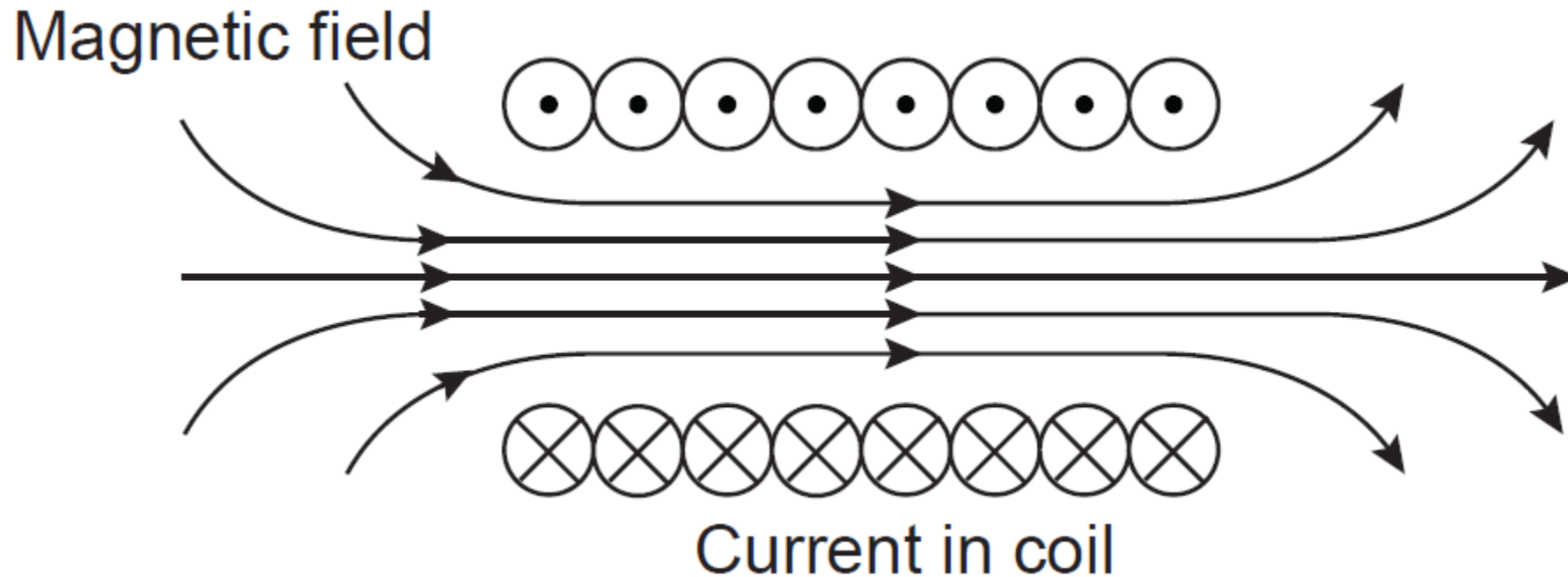
- Acceleration is provided by RF field instead of DC bias
- Variation in field gives time dependence of emitted current
- Naturally produces “bunches”
- Integrates well into additional acceleration stages



Solenoid focusing

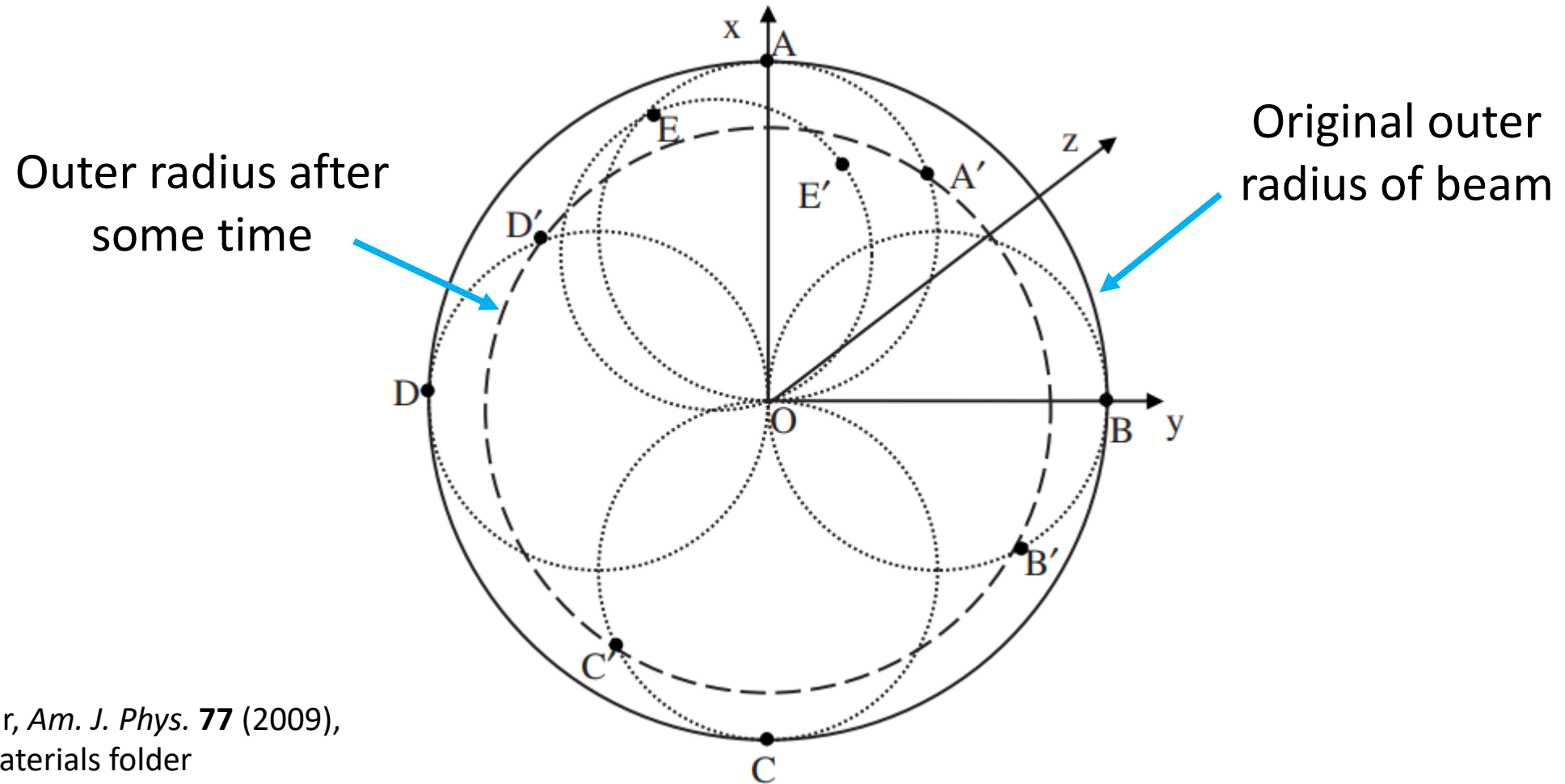
Solenoid focusing

- Solenoid focusing relies on fringe of magnetic field at entrance/exit
- Sketch of a solenoid magnet and field:



Solenoid focusing

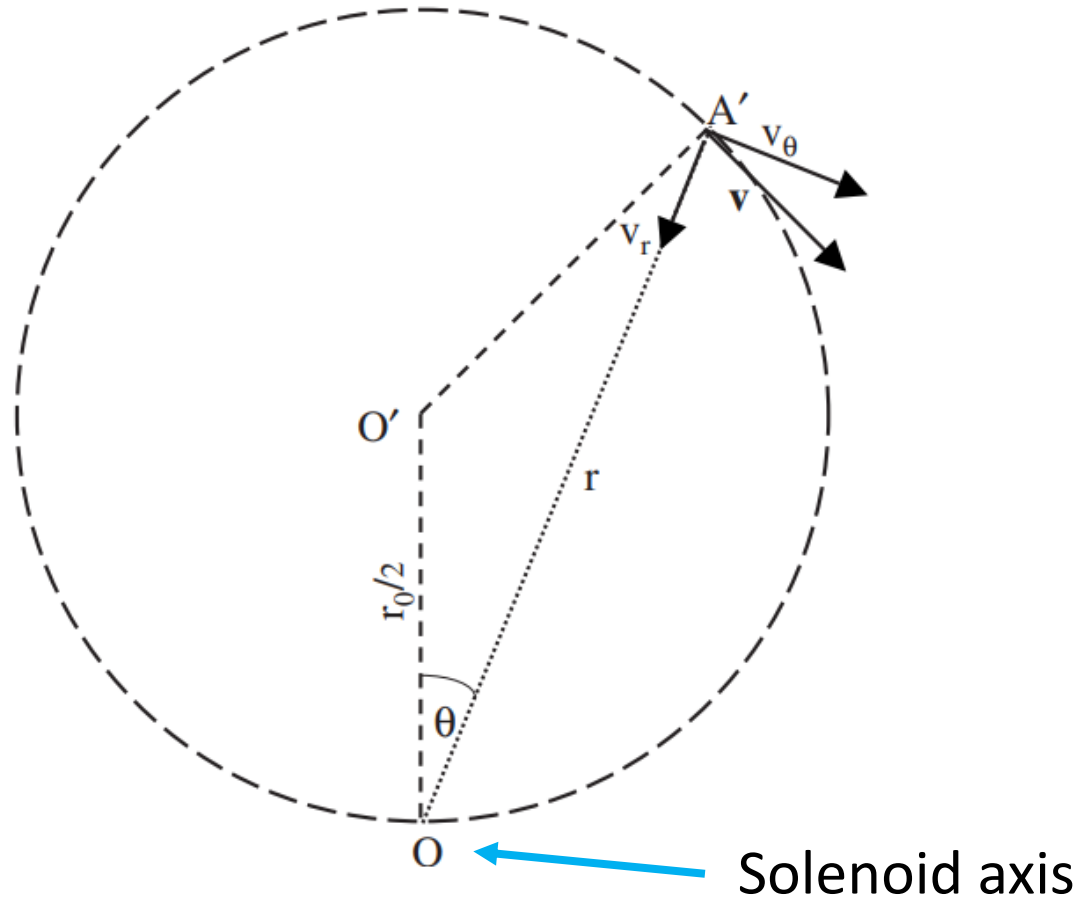
- Example orbits of particles going through the magnet:



See V. Kumar, *Am. J. Phys.* **77** (2009),
in lecture materials folder

Combining emission types

- Radial component of velocity:



See V. Kumar, *Am. J. Phys.* **77** (2009),
in lecture materials folder

Solenoid transfer matrix

General, note it is the same for both transverse coordinates

$$\mathcal{M}_s(z_0|z) = \begin{pmatrix} \cos \phi & \frac{2}{S} \sin \phi & 0 & 0 \\ -\frac{S}{2} \sin \phi & \cos \phi & 0 & 0 \\ 0 & 0 & \cos \phi & \frac{2}{S} \sin \phi \\ 0 & 0 & -\frac{S}{2} \sin \phi & \cos \phi \end{pmatrix}$$